

aTOMos

v3.6.1

Total Integrated Engine

Foundational knowledge substrate

One integrated state. Versioned profiles.

Coherent batches. Explicit operator domains.

The OTAN2 invariant bank inside the engine.

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Author-supplied date	10-07-1990
Engine family	atomOS v3.6
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Master / substrate	M2 / S1
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Full master specification and accompanying reference

Identity and double UU; Ψ and lineage; log-polar space and topology; RK4; JK; ASA/NA; blend; OTAN2; invariants; coherent inputs; field, matrix and transfer interfaces; atomic epochs; provenance; storage; future-variant contracts and blue-team tests.

The new Markdown attachment is integrated throughout the master and mapped into a versioned substrate profile. Editable sources, a complete local mathematical fixture, a Python reference and reproducible tests accompany this PDF.

Release record

aTOMos v3.6.1 is the new integrated release within the atomOS v3.6 family. This volume restates the complete engine and incorporates the attached profile proposal into its configuration, input, numerical-model and control-policy interfaces. It is a master, not a separate aperture appendix or panel-only document.

Release field	Definition
Author	Tom Klootwijk; NL200678942; 10-07-1990, as supplied.
Version	Family v3.6; release 3.6.1; master M2; substrate S1.
Foundation	Established M1 core plus the new attachment's named extension requirements.
Source supplied now	Pasted <code>markdown.md</code> , 152 lines; retained exactly in the ZIP.
Profile registry	A foundational declaration and a separate, fully populated mathematical reference fixture.
Executable reference	Local preprocessed word epochs, explicit JK events, OTAN2, invariant checks, coherent records and replay hashes.
Publication package	This PDF, editable $\text{\LaTeX}$, Python modules, JSON schemas/profiles, source map, examples and tests.

What is supplied and what is declared

The attachment names the B-profile, actual eigenmatrix, optical/phase transfer, sampling substrate, dimensional adapter and J/K feedback policy, but supplies no numerical calibration or concrete physical feedback equation. Its wording is preserved as a *source proposal*; the master fills those slots with typed declarations and source status, not invented measurements. The local reference profile fills its own mathematical test values separately. [D, lines 124–148]

A future variant can bind a declared slot by naming the actual formula, parameters, units, frame and source. Its parent profile remains identified by digest. This makes S1 a stable substrate for future variants without treating a not-yet-provided physical model as executable data.

Local execution contract

The supplied package has no remote activation, license check, telemetry service, use-case classifier or shutdown endpoint. Its finite-number, type, dimension and coherence checks implement the mathematical input domain. Invalid input produces an ordinary error, leaves committed state unchanged and does not create a persistent lockout. These are different mechanisms.

The new code is a local reference implementation, not a patched vendor driver or a recovered earlier binary. A concrete acquisition or actuator interface is not present in the source code delivered here. Capability and test coverage are listed explicitly near the end of the volume.

Source layers and the attachment's structure

A is the 91-page WHITE KING corpus, **F** the eight-page fly-eye addendum, **B** the five-page blend source, **O** the seven-page OTAN2 source and **D** the newly attached Markdown. **M1** denotes the prior integrated mathematical contract established in the project conversation. **N01–N10** identify current completion decisions.

The M1 PDF itself is not mounted in this preparation environment. The opaque reference pdf_v9BtD7.pdf inside D is not a separately available document either. The master therefore uses D's quoted relationships as supplied text, not as proof that its reported section numbers match a separately inspected file. This volume has its own section references. [N01]

D lines	Attachment sequence	Master integration
1–3	Localized spatial aperture processor	Named preprocessed-input profile.
25–29	Configuration snapshot and envelopes	Typed boundary and geometry bindings.
31–35	Faceted/emitter input link	Explicit coherent record batches.
37–43	Future JK feedback profile	Policy port; no guessed mapping.
45–50	Temporal coherence and type precedence	Batch/domain and predicate-resolution rules.
74–88	Field f , B-profile and eigenmatrix	Distinct evolution and reference-matrix obligations.
90–96	Sampling substrate and chart increments	Representation/transfer layer; source claim retained separately from OTAN2 semantics.
124–142	Missing profiles and dimensional restrictions	Complete obligation registry and conformance checklist.
146–148	New named engine profile	S1 parent/variant manifest.

Supplied text is not silently converted into a missing implementation.

A declaration says what a slot must contain. A bound profile supplies that content. A worked fixture supplies test values. An executed test reports finite evidence. These four levels are distinct fields in this master and the ZIP.

The new attachment's repeated AI-export notices, interface prompts and literal svg lines remain in the archived original. They are not operator definitions. Its proposed physics and feedback interfaces are incorporated without importing those user-interface fragments into the mathematical core.

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I / THE TOTAL ENGINE

1 One integrated state and transition

The master transition is

$$\mathcal{E}_{3.6.1, \Theta_n} : (\mathcal{S}_n, I_n) \mapsto (\mathcal{S}_{n+1}, R_n). \quad (1)$$

Its state is the product

$$\mathcal{S}_n = \langle U, C_n, X_n, Q_n, \Theta_n, D_n, L_n \rangle. \quad (2)$$

Field	Role	Contents
U	Identity registry	Full digests, author attribution and symbolic genesis records.
C_n	Optional procedural state	Lineage capsules, declared coordinates and producer statuses.
X_n	Logical words	Occupancy or explicitly named predicate support.
Q_n	JK state	One bit or an indexed bank of stored bits.
Θ_n	Snapshot/profile	Dictionary, masks, fields, frame, transfer and selected policies.
D_n	Diagnostics	Typed blend/predicates, OTAN2 views and invariant-bank results.
L_n	Provenance	Ordered epoch records and optional versioned hash-chain state.

[A, pp. 11–14, 16–18][M1, product-state contract]

The new input substrate is an extension of I_n and Θ_n , not a replacement for the word or JK algebra. A batch chooses a producer representation; it does not make a raw antenna sample, a coordinate index and a bitmask interchangeable.

1.1 Independent computations, one committed record

The epoch computes ASA/NA outputs, explicit JK transitions, declared blend results and orientation diagnostics from a consistent pre-epoch snapshot. Their result fields coexist in R_n . When one diagnostic is undefined, that fact need not reset an otherwise defined word or JK bit.

The reference implementation realizes the `local_preprocessed_words/1` profile. Its state contains the next epoch number, stored word outputs, JK bit, last committed interval end and ledger head. The other source layers are restated in this master and exposed through separately tested utilities or declared producer ports. That coverage boundary is explicit rather than hidden behind a claim that every profile has been executed.

2 The versioned foundational substrate

S1 packages the inherited core contract and the new extension declarations into a profile registry:

$$\Theta = \langle \text{version}, \text{parent}, \text{dictionary}, \text{geometry}, \text{evolution}, \text{transfer}, \text{policy}, \text{provenance} \rangle. \quad (3)$$

A field's status is `bound`, `declared_unbound`, or `not_applicable` for a named profile. A fixture value additionally carries `authored_test_fixture`. These status labels describe knowledge and configuration; they are not user authorization levels. [N02]

2.1 Two profiles in the ZIP

Foundational declaration. `profiles/foundation.json` names every slot in `D`, its source location, expected data and relation to the master. It is the parent contract for later variants.

Local mathematical reference. `profiles/local_reference.json` binds a finite word dictionary, explicit masks, a simple index conversion, a test matrix, a constant-field RK4 fixture and explicit JK events. It is fully populated for its selected local model. It does not label those test values as hardware coefficients supplied by `D`.

2.2 Version inheritance

A derived profile identifies its parent by content digest, lists replaced fields and tags each new value's source. The old profile remains available for reproducing old epochs. A derived version cannot silently change the meaning of the same mask, frame or coefficient while retaining the parent's identity.

Proposition 2.1 (Profile identity separation). *If a profile change modifies any canonically represented value, it changes the canonical input to its profile digest. A run records both the human-readable profile ID and that content digest.*

Proof. The serializer preserves field names, types, array order and values. A changed value therefore changes the serialized record. The digest is computed from those bytes and its domain tag; hash equality is not used as a mathematical guarantee against every possible collision. $\square$

The attachment says future policies are new named engine profiles. S1 formalizes that as an ordinary data and provenance relationship. It does not infer permission from an author's identifier or install a compulsory remote profile registry. [D, lines 37–43, 146–148]

3 Names, types and source roles

Name	Master field	Meaning
Ψ	State or tagged finite word	The full record and a register specialization remain distinguishable.
ϕ	Chart phase / named hinge	Angular coordinate, not automatically an eigenaxis orientation.
A	A_{axis}	Source reference axis; distinct from the ASA mask.
W	Interval	Common duration for the components that claim to share it.
B_{ϕ}	Kinematic descriptor	Earlier double-lowercase-phi role. D's proposed medium profile is separately qualified.
N	N_{pinion}	Datum/reference role; distinct from NA.
ASA / NA	$M_{\text{ASA}}, M_{\text{NA}}$	Selector masks in fixed stage order.
Boundary	M_{∂}	Blocking predicate, with whole-word disposition.
V	$\mathcal{V}$ / valid mask	Gestation/admission vocabulary versus finite valid-bit extent.
Jitter	$\mathcal{J}$	A supplied word or procedural perturbation, not JK input J.
JK	$J_{\text{JK}}, K_{\text{JK}}, Q$	Two supplied bits and persistent logical state.
OTAN2	β, e	Log-increment orientation and its relative angle.
Eigenmatrix	A_{ref}	Declared matrix defining a reference; separate from a proposed aperture/beamforming coefficient array.
Coherent batch	I_n	Same declared clock/frame/profile and explicit interval.

[B, pp. 1–5][D, lines 124–142][N03]

D describes the B-profile as medium physics, while the earlier blend source used B for a kinematic calculus. The master retains both source meanings as `B_kinematic` and `B_medium` roles. They can be related by a supplied application profile; they are not automatically the same numerical object.

A binary representation does not erase a type. An identifier, coordinate key, occupancy bit, sampled value and latch input still have different contracts after being stored in machine words. This distinction is the basis of the future-variant substrate.

4 Genesis and double UU

The retained double-UU relation is

$$\text{UU}(C) = \langle \iota, \text{StateRecord}(C) \rangle, \quad \iota \in \{0, 1\}^{256}. \quad (4)$$

It binds a full identity digest to its state record. The author's supplied name, identifier and date are attribution metadata. They are not a numerical proof of location or an executable privilege. [A, pp. 16–18][M1, identity contract]

The source's corrected named genesis union is retained as a documentary label set. The worked reference uses author-symbol and asa-symbol, keeping person-derived coordinates out of its numeric fixture. The set union and the ordered identity/state pair are different constructions. [A, pp. 75–78]

4.1 Versioned symbolic genesis

The supplied reference normalizes symbolic labels to Unicode NFC, takes unique members, sorts by UTF-8 byte order and hashes a tagged canonical record with a named domain and length:

$$\iota = \text{SHA256}(d_G \parallel \text{len64}_{BE}(C(G)) \parallel C(G)). \quad (5)$$

The exact profile is recorded as aTOMos-genesis-3.6.1. It is not claimed to reproduce historical digests generated under different encodings or domain strings.

Proposition 4.1 (One-bit projection cannot replace the identity). *There is no injective representation of all 2^{256} digest values in one bit. A smaller finite coordinate dictionary likewise cannot retain every possible identity without an independent registry.*

Proof. The output domain has fewer values than the input domain. At least two distinct possible inputs must share an output. Keeping a bit flag or a finite placement index therefore cannot remove the need to retain identity separately. $\square$

A synthetic hash-to-chart placement may split a digest into two 128-bit integers and map them into declared coordinate ranges. Such a projection is a placement profile, not a measurement model. The core and the new coherent-batch layer retain the full identifier and do not re-label a hash mutation as a measured trajectory change.

5 The original word and lineage core

The source production vocabulary remains

$$0 \rightarrow 0, \quad 1 \rightarrow 1[\phi_+]1[\phi_-]. \quad (6)$$

The source also specifies fixed-width shifts and both XOR and OR merges. The master keeps the merge choice explicit:

$$S_{\oplus}(\psi) = ((\psi \ll 1) \bmod 2^b) \oplus (\psi \gg 1), \quad (7)$$

$$S_{\vee}(\psi) = ((\psi \ll 1) \bmod 2^b) \vee (\psi \gg 1), \quad (8)$$

$$\psi' = S_{\mu}(\psi) \oplus \mathcal{J}. \quad (9)$$

[A, pp. 9–14]

A declared zero-absorbing grammar applies its source-zero guard before jitter; otherwise a nonzero jitter word can change zero. The reference exposes that choice rather than silently assuming unguarded XOR preserves $0 \rightarrow 0$. With $\psi = 5$, sufficient width and zero jitter, the XOR merge gives 8 and the OR merge gives 10.

5.1 Lineage addresses are not the literal register shifts

The retained record convention is

$$\ell_b = 2\ell + b, \quad b \in \{0, 1\}, \quad \text{parent}(\ell_b) = \ell_b \gg 1, \quad (10)$$

with root one. A child address consists of the leading root bit followed by the branch sequence. Therefore distinct branch paths have distinct addresses until the declared finite width is exceeded. Overflow is reported before a new address is formed.

The grammar, the path identity and the geometric location are three fields. Two lineages can occupy the same logical cell without becoming the same identity. Occupancy union may merge their support while the registry retains their provenance.

5.2 Master connection

A word epoch can consume supplied preprocessed words directly; it need not execute a branching producer. A selected procedural variant must declare how proposals become dictionary words and whether the source hinge changes position or direction. This producer contract is part of Θ , not inferred from the choice of a bitwise operator. [M1, producer profiles]

6 Log-polar space, topology and memory identity

Use a fixed positive reference radius:

$$\rho = \ln(r/r_*), \quad r > 0, \quad \phi \in \mathbb{R}/(2\pi\mathbb{Z}). \quad (11)$$

The planar display map is $X = r_* e^{\rho}(\cos \phi, \sin \phi)$. The source alias $\theta \equiv \phi$ names the selected single-angular-coordinate profile; a general independent 3D angle is not stored in that same two-scalar record. [A, pp. 22–26][D, line 141]

6.1 Quantization is a profile choice

The existing finite dictionary uses radial rows and periodic angular entries. Floor bins and nearest periodic nodes are different conventions; the profile must name its choice. The executable reference processes supplied logical words and calibrated increments, so it does not silently infer a physical cell from a scalar integer key.

6.2 The retained quotient and mirror

A declared Klein-chart profile uses

$$(u + 1, v) \sim (u, v), \quad (u, v + 1) \sim (-u, v). \quad (12)$$

For $k = \lfloor v \rfloor$, the representative is

$$\bar{v} = v - k, \quad \bar{u} = \text{frac}((-1)^k u). \quad (13)$$

The tangent and any field component crossing the seam transform by the corresponding Jacobian. A mirror is not automatically a field symmetry: it requires compatible data and an update commuting with the selected reflection.

On P angular nodes, $M(j) = (-j) \bmod P$ and $T_\delta(j) = (j + \delta) \bmod P$ give a mirror involution and an inverse portal pair. Whole-word disposition additionally requires a geometry permutation to preserve complete logical groups before any absorption-equivalence claim is made. [M1, topology, mirror and portal profiles]

The reference exports the quotient representative as a pure utility. The normal coherent word epoch uses an already declared dictionary/frame and does not silently fold a raw 3D observation into that quotient.

7 RK4 and the evolution interface

The source's four-stage numerical core is retained for a supplied first-order field f :

$$k_1 = f(t, z), \quad k_2 = f(t + h/2, z + hk_1/2), \quad (14)$$

$$k_3 = f(t + h/2, z + hk_2/2), \quad k_4 = f(t + h, z + hk_3), \quad (15)$$

$$z^+ = z + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (16)$$

The state dimension, component units, time unit, step and substeps are part of the field profile. A failed field dimension or a nonfinite intermediate does not become a valid next state. [A, pp. 19–24][D, lines 74–80]

7.1 What the new attachment requests

D proposes using this slot for propagation, Doppler or phase-evolution calculations. It does not provide a right-hand side, medium parameters, initial conditions or boundary values. The master retains those as the declared evolution binding in S1.

A spatial partial differential equation is not itself a two-component time derivative. To use a first-order numeric kernel, a supplied profile must describe the chosen state, any spatial representation and the reduction to a first-order evolution on that state. The original chart coordinates can then be derived or observed through a separately declared map. This is an interface requirement, not a selected wave solver. [N04]

7.2 Complete local fixture

The reference test profile uses the authored constant field

$$f_{\text{fixture}}(t, z) = (0.25, 0.5) \quad (17)$$

with time unit `test_tick`. Starting at $(1, 2)$ and taking $h = 0.125$ yields exactly $(1.03125, 2.0625)$ in the declared floating test. This is a conformance fixture for RK4 and chart types. It is not a measured material model or an antenna wave equation.

7.3 Sampling and disposition

Checking an evolving path only at substep endpoints is a declared sampled policy, not continuous boundary detection. A numerical producer must record that schedule and its dictionary compiler. The current local word epoch instead receives preprocessed words; it does not infer a trajectory between those words or execute an unspecified propagation field.

8 Field quantities and dimensions

The earlier source includes

$$\nabla \cdot u = \frac{1}{r} (\partial_\rho u_r + u_r + \partial_\phi u_\phi), \quad (18)$$

when u_r, u_ϕ are physical orthonormal planar components and $r = r_* e^\rho$. For coordinate rates $a = \dot{\rho} = u_r/r$ and $b = \dot{\phi} = u_\phi/r$, the equivalent form is

$$\nabla \cdot u = \partial_\rho a + 2a + \partial_\phi b. \quad (19)$$

The component convention matters. Neither expression is applied directly to an unsigned occupancy word. [A, pp.30–33][M1, field typing]

D introduces medium properties as a B-profile obligation. These properties need their own unit-tagged record, equations and constitutive relationships. Naming a velocity, density or permeability does not supply the remainder of the model, its geometry or its numerical state. [D, line 128]

8.1 Physical state versus chart observation

Let $u_n \in \mathcal{U}_\Theta$ denote a supplied field state and let C_Θ denote its declared chart observation. A profile can name

$$u_{n+1} = \mathcal{R}_\Theta(u_n), \quad z_n = C_\Theta(u_n), \quad (20)$$

without identifying the field state with the two chart coordinates. The types, dimensions and implementation of $\mathcal{R}_\Theta$ and C_Θ belong to that profile. S1 records the obligation; it does not choose their unprovided coefficients.

8.2 Count is not geometric measure

A population count counts bits. A planar annular-bin area instead has the form

$$A_{ij} = \frac{1}{2}(r_{i+1}^2 - r_i^2) \Delta\phi. \quad (21)$$

A weighted measure requires those weights as separate data. A one-bit field cannot retain a discarded continuous distance, density or gradient merely because its source called it an SDF. The fixed core keeps the binary predicate and any continuous quantity in different records.

II / LOGIC, APERTURE AND BLEND

9 The J–K state bank

The exact binary transition is

$$Q^+ = (J \wedge \neg Q) \vee (\neg K \wedge Q), \quad Q, J, K \in \{0, 1\}. \quad (22)$$

J	K	Q^+	Mode
0	0	Q	Hold
1	0	1	Set
0	1	0	Reset
1	1	$1 - Q$	Toggle

The source supplies JK set and toggle language; the complete Boolean model is retained from the prior integrated engine. It is not renamed as an inferred biological effect or device command. [A, pp. 53–55][M1, JK contract]

9.1 Sequential events in a coherent epoch

The reference accepts an explicit, time-ordered list of (J, K) events. Each event reads the preceding logical Q and produces the next. Events with equal timestamps preserve their supplied list order. With no event the state holds.

A set followed by a toggle ends at zero, whereas a toggle followed by a set ends at one. Therefore the event list cannot be replaced by an unordered set even though every individual J/K pair is legal. Event ordering is part of the input semantics.

9.2 The proposed feedback extension

D suggests a future policy consuming a typed observation and producing a J/K pair. It supplies no actual map, thresholds, state dependence or update schedule. S1 therefore declares

$$\pi_{JK} : \mathcal{D}_\pi \times \mathcal{Q}_\pi \longrightarrow \{0, 1\}^2 \quad (23)$$

as a named policy interface, not a numerical policy already implemented by the source. [D, lines 37–43, 133]

The complete local fixture uses `explicit_events`. It exercises all four table modes without using an angular value as an implicit trigger. An undefined OTAN2 result remains a diagnostic status and leaves those supplied events unchanged.

A future policy can be represented as a different named variant with its own domain and tests. It must not silently replace explicit-event semantics while keeping the same profile identity. This is the precise role of the policy slot within the integrated engine.

10 The ASA/NA four-word kernel

For shared logical word support, the kernel is

$$a = x \wedge M_{\text{ASA}} \wedge M_{\text{valid}} \wedge M_{\text{fringe}}^{\text{eff}} \quad (24)$$

$$n = a \wedge M_{\text{NA}}, \quad (25)$$

$$h = \text{popcount}(n \wedge M_{\partial}), \quad (26)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases} \quad (27)$$

The result is $F_{\text{ASA}} = (a, n, h, y)$. In record mode $M_{\text{fringe}}^{\text{eff}} = M_{\text{valid}}$; in explicit-fringe mode it is the complete supplied fringe mask. [A, pp.75–81][M1, ASA core and fringe extension]

10.1 The source order is retained

The first selection is ASA, the next is NA, and the intersection count is evaluated after both. The source's >0< symbol names the paired-wing/central-aperture stage. The NA-after-ASA order is not collapsed even though the Boolean intersections themselves commute, because the intermediate record identifies each stage's contribution.

ASA, NA and fringe one-bits select support. Boundary one-bits block selected support. Those polarities are different. A meaningful bitwise AND requires that every operand refer to the same dictionary and grouping.

10.2 Whole-word disposition

If one selected bit intersects the boundary, the complete selected word becomes zero. The alternative $n \wedge \neg M_{\partial}$ would clear only individual boundary bits and is not this operator. The reference's test uses $n = 255$ and boundary 128: the output is zero, not 127.

The last angular word is restricted by its valid-bit mask. Padded columns and unused high bits cannot become observations. The implementation additionally checks that stored planes and incoming occupancy contain no nonzero tail bits; it does not silently truncate an invalid source record.

10.3 Admitted state and retained record

The source's V gestation/admission role is retained as the distinction between proposed support and emitted y . The preceding fields and disposition remain in the epoch record even when $y = 0$. Zero output does not erase the identity or the evidence of the transition. [A, pp.6–7, 33–41]

11 Kernel laws and backward recovery

Proposition 11.1 (Selector idempotence and contraction). *For fixed M , $P_M(x) = x \wedge M$ obeys $P_M(P_M(x)) = P_M(x)$ and never creates a one-bit absent from x .*

Proof. At each bit, applying the same Boolean intersection twice has the same truth value as once. A one-bit in the result requires a one-bit in the input. $\square$

Theorem 11.2 (Final-word idempotence). *For fixed masks and one selected profile, reprocessing the final output word yields that same final output.*

Proof. An absorbed output is zero and stays zero. Otherwise the output already satisfies each selector and has no boundary intersection. The same selectors leave it unchanged and the same boundary check remains clear. $\square$

11.1 Neutral fringe

With $M_{\text{fringe}} = M_{\text{valid}}$, the new effective aperture gives exactly the inherited ASA input because the valid mask is idempotent. Every later stage therefore has the same input. This recovers all four fields, not merely the final population count.

11.2 The preserved non-monotonicity

Take

$$x = 3, \quad M_{\text{ASA}} = M_{\text{NA}} = M_{\text{valid}} = 3, \quad M_{\partial} = 2. \quad (28)$$

The unmodified output is zero. Selecting fringe 1 removes the blocking bit before the group test, producing output 1. A narrower selector has restored a group that was previously absorbed. Consequently the final output is not necessarily monotone under selector inclusion.

This counterexample is part of the master compatibility contract and the new regression suite. Replacing whole-word disposition with a monotone filter would change the engine even if many ordinary examples still looked similar. [M1, aperture laws]

11.3 Diagnostics do not implicitly change the core

Under the reference's explicit-event/record policy, changing an OTAN2 reference axis changes its residual but does not alter the core masks, grouped output or J/K events. This projection identity is tested by comparing complete records before and after that isolated change. New feedback variants must state where that identity is intentionally replaced.

12 A–W–B–N and the three predicate roles

The literal source expression is

$$s = N \oplus (A + \phi(W)) \pmod{2}. \quad (29)$$

A declared binary specialization supplies $n_b, a_b, \beta_b \in \{0, 1\}$ and computes

$$s_{\text{blend}} = n_b \oplus a_b \oplus \beta_b. \quad (30)$$

All eight bit triples are defined. The kinematic contribution bit is an encoder output, not the duration W or an entire continuous matrix. [B, pp. 1–2]

12.1 One spelling, several source roles

The source’s “double lowercase phi” remains a descriptor name. Its displayed single ϕ does not select a unique second derivative, a squared phase or a transformation matrix. D adds a proposed medium-physics meaning to a B-profile; that role is recorded separately from the original kinematic descriptor. A profile can bind a relationship only by supplying it.

Predicate	Meaning of zero	Meaning of one
s_{align}	Encoded alignment/no phase adjustment	Discrepancy or adjustment condition
s_{limit}	Inside the declared envelope	Limit/threshold event
s_{fringe}	Classified hollow region	Classified active fringe

[B, pp. 2–5]

These flags are separately resolved *before* the whole-word selection stage, as requested in D. Resolution can be a defined Boolean or an explicit unspecified status. It does not mean that all three flags acquire the same value, or that an unprovided alignment encoder becomes a true acceptance gate. [D, lines 49–50][N05]

The current word profile records the flags but does not use them to rewrite selectors. A selected profile that consumes a flag must specify its map and polarity. This preserves the original bit semantics while giving the new attachment a precise place in the execution sequence.

13 Aperture boundaries, hollow region and fringe

The source describes a hollow center and an active surrounding fringe, and mentions several possible configurations. It does not supply a selected geometric system, numerical inner/outer boundary or a complete field response. [B, pp. 3–5][D, line 29]

A bound geometry profile contains

$$G = \langle \text{kind}, \text{frame}, b_{\text{inner}}, b_{\text{outer}}, \text{units}, \text{domain}, \text{edge rule} \rangle. \quad (31)$$

The bounds can have different mathematical forms in different profiles: angular bounds, radial surfaces or explicitly named sets. A radius is not changed into an angle by using the same field label. Unit and frame tags are therefore required.

13.1 Binary region interpretation

For a supplied classification with hollow set H and active set R ,

$$s_{\text{fringe}}(z) = 0 \quad (z \in H), \quad s_{\text{fringe}}(z) = 1 \quad (z \in R), \quad (32)$$

with $H \cap R = \emptyset$. The outer domain and exact-edge rule must be declared. A point outside both sets is not automatically an active-fringe hit.

13.2 A complete mask versus a geometric derivation

The local test profile supplies mask words directly. It therefore has a complete Boolean aperture at the chosen dictionary, without pretending that those words were derived from a Cassegrain, Gregorian or phased-array physical model. Its geometric-source status is explicit in the profile.

For a later geometry profile, a mask compiler maps its classified domain into the dictionary. The compiler version and the full geometry binding become part of the profile snapshot. Recompiling with another edge convention changes that snapshot even when the bit count coincidentally remains the same.

D's claim that the listed hardware parameters can populate the envelope fields is retained as the intended extension architecture. It does not determine their actual values. The foundational profile records the alternatives and the required binding, while the worked profile uses a separately labelled Boolean fixture.

III / THE ATTACHED SUBSTRATE, FULLY MAPPED

14 The coherent input batch

D explicitly requires not mixing clock frames and resolving types before final word selection. The new concrete batch contract is

$$I_n = \langle n, b, c, f, d, \theta, [t_0, t_1), \mathcal{W}, \mathcal{J} \rangle, \quad (33)$$

where b is batch ID, c clock ID, f frame ID, d dictionary ID, θ profile digest, $\mathcal{W}$ logical word records and $\mathcal{J}$ ordered JK events. [D, lines 31–50, 90–96, 142][N05]

Each word record repeats the batch, clock, frame and profile identity and supplies a timestamp $t \in [t_0, t_1)$. Exactly one record exists for each required logical word. The fixed snapshot provides masks; the record supplies its occupancy and optional encoded/angle observations.

14.1 Logical coherence is not a phase measurement

Two records carrying the same clock string are *declared* to share the batch's clock domain. That label alone does not prove a physically synchronized oscillator, known phase offset or calibrated clock conversion. A real acquisition profile needs those definitions separately. The current batch checks establish consistency of the supplied records.

D's instruction not to mix timelines is implemented as equality checks, interval membership and explicit epoch ordering. Records from another clock or snapshot are not silently renamed to match. A later resampling or clock-conversion adapter would need a named transformation with provenance; none is inferred here.

14.2 Source order and canonical output

Records may arrive in a different list order, but output words are emitted in canonical logical order. The exact input ordering remains represented in the input digest. Thus semantic word equality and source-byte/order identity are separate checks.

The implementation preserves the original batch and configuration by copying them before evaluation. A rejected batch leaves the state and stored event list unchanged. The next valid batch can run normally; rejection is not a persistent engine lock.

15 Time intervals, epoch order and numerical values

The local reference represents batch and event times as integer nanoseconds in the declared clock domain. The selected interval is half-open, so t_0 is included and t_1 is excluded. This convention avoids assigning an endpoint to two adjacent batches. It is a new concrete record choice, not a timestamp unit supplied by D. [N05]

15.1 Epoch identity

The input epoch equals the engine's next epoch counter. A subsequent committed batch starts no earlier than the prior interval's end. Replaying an already committed batch into the live reference raises an ordinary input error. Deterministic replay is performed in a fresh state with the same profile and inputs, not by mutating the old epoch counter silently.

A timestamp inside a record does not establish acquisition latency, clock accuracy or real-time throughput. Those require additional measured metadata. The timestamp domain exists to make the input relation explicit.

15.2 Finite numeric representations

The reference accepts finite real-valued chart increments and finite or explicitly absent axis values. A nonfinite JSON value is rejected as a malformed mandatory numeric input. A finite input at a singular OTAN2 point instead produces its defined diagnostic status. Those are different paths.

Integer word values are genuinely integers, not Boolean values or floating approximations. In Python, Boolean is an integer subclass; the validator deliberately checks the exact type so that a Boolean does not accidentally pass as a 32-bit occupancy word.

15.3 Two historical rates remain distinct

The older corpus names 128 ticks and the fly-eye text names a 250 Hz example. The master keeps the period calculation

$$T_{128} = 7.8125 \text{ ms}, \quad T_{250} = 4 \text{ ms}, \quad T_{\text{common}} = 0.5 \text{ s}. \quad (34)$$

That common interval contains 64 and 125 events, respectively. The local fixture's 125,000,000 ns interval is an authored batch example and is not claimed to select either physical device rate. [A, p. 83][F, p. 7]

16 The B-profile and propagation obligation

D says a physical profile should supply medium-dependent propagation and an explicit field $f(t, z; \Theta_n)$. Its text names examples but contains no actual differential equations or parameter values. S1 records the following required binding:

$$B_{\text{medium}} = \langle \text{medium, state, units, equations, coefficients, initial/boundary data, discretization} \rangle. \quad (35)$$

[D, lines 74–88, 128]

16.1 Separate the three mathematical roles

Evolution changes a supplied state under a chosen time law. **Observation** maps that state or preprocessed samples to chart quantities. **Coordinate conversion** changes representation or units. A Doppler-derived observation, for example, cannot be treated as the complete time derivative merely because both appear in one processing chain.

The phrase “append the physics to RK4” is therefore interpreted as a requirement to bind an evolution-compatible state and right-hand side. It is not an implementation of every physical law named in the attachment. A spatial field representation must be declared before a time-stepping method can act on it.

16.2 Completeness record

For this source, the status of the physical B-profile is `declared_unbound`. Its expected fields are populated in `foundation.json`; no acoustic velocity, density, permeability or frequency is guessed. That status does not prevent a profile which operates only on already supplied words and chart increments from being a complete local computation.

The fixture’s constant two-component field is separately available for RK4 conformance. Its code path and units are explicitly labelled as an abstract numerical example. It never changes its provenance tag to “measured medium” merely because the test passes.

16.3 Future binding rule

A variant which supplies an actual B-profile records its equation source, units, state dimension, representation and tests. If that profile changes the dimension of the evolving state, its chart observation adapter must change with it. The two-coordinate core is not silently enlarged by dropping an extra component into the same record.

17 The eigenmatrix and reference-axis obligation

D uses “actual eigenmatrix” for array coefficients and relates it to directional processing. The earlier OTAN2 source instead describes a preferred local direction of a transformation matrix. These roles are not automatically the same matrix. [D, lines 88, 129][O, p. 7]

S1 separates

$$A_{\text{ref}} \in \mathbb{R}^{m \times m} \quad \text{and} \quad W_{\text{aperture}} \in \mathbb{C}^{p \times q} \quad (36)$$

as type declarations only. The real/complex field and dimensions are examples of what a binding must specify, not asserted source dimensions. A chosen profile records its actual coefficient field, basis, units and role. [N03]

17.1 Reference construction

A reference-axis binding identifies its matrix or constructor, the eigenpair selection rule, degeneracy handling, normalization and sign/direction convention. If an eigenvalue is repeated, an arbitrary numerical eigenvector is not a uniquely specified direction. If the reference is an unoriented line, the meaningful angular class differs from a directed vector.

The current word epoch accepts an explicit α already expressed in the log-tangent frame. It does not compute array weights or estimate an incoming direction. This is a complete input contract for OTAN2 even while the proposed upstream matrix is unbound.

17.2 Complete matrix fixture

The reference profile includes

$$A_{\text{test}} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad v_{\text{test}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \lambda_{\text{test}} = 2, \quad (37)$$

so $A_{\text{test}}v_{\text{test}} = 2v_{\text{test}}$. This is an explicit eigenpair identity fixture in a test basis. It is not an array beamforming matrix or a calibration inferred from the hardware names in D.

The master retains the source’s moving-axis wording while making its numerical obligation precise. An unprovided matrix remains unprovided; its downstream explicit-angle interface still has exact, testable semantics.

18 The optical/phase transfer interface

The actual coordinate conversion written in D is

$$\Delta\rho = h_\rho\Delta i, \quad \Delta\phi = h_\phi\Delta j. \quad (38)$$

This maps preprocessed index changes to chart-unit increments. It does not, by itself, define an analog receiver transfer function, a phase-unwrapping algorithm or a raw signal estimator. [D, lines 130–132]

18.1 Versioned transfer record

A transfer binding records input/output units, frame IDs, spacings, offset conventions, sign conventions, calibration provenance and its phase-lift policy. The reference's implemented `index_scale/1` uses finite positive h_ρ, h_ϕ and retains the calibrated pair in every OTAN2 record.

The worked spacings are

$$h_\rho = \frac{1}{8}, \quad h_\phi = \frac{1}{16} \text{ rad}, \quad (\Delta i, \Delta j) = (4, 8), \quad (39)$$

which give $(\Delta\rho, \Delta\phi) = (0.5, 0.5)$. Doubling both index resolutions and halving both spacings gives the same represented increment. The tests check that representation identity rather than using uncalibrated raw indices.

18.2 No automatic analog-to-coordinate reduction

A physically bound optical or phase transfer table needs actual input quantities and a defined model. D does not supply its entries. S1 therefore keeps `optical_phase_transfer` as the full interface and `index_scale/1` as one deliberately narrower implemented representation conversion.

A real sensor phase is not automatically the spatial chart phase. An increment in a sample register is not automatically a radial displacement. Such identifications require the supplied adapter's meaning and calibration, not a matching variable name.

18.3 Mask and coordinate roles

A transfer result may later be classified by a geometry profile, but Eq. (38) does not itself choose a bitmask or an absorption grouping. The word core still expects words in the declared dictionary. The reference links preprocessed increments to diagnostics while keeping the supplied occupancy words as independent typed inputs.

19 Sampling, dimensional reduction and the DOA claim

D names a Dirac-comb sampling operator, convolution and a Fourier substrate. It supplies no sample period, filter coefficients, bandwidth, array delays or transform normalization. The master records these as a *sampling-interface obligation*; it does not manufacture a raw-acquisition algorithm from those names. [D, lines 90–96, 142]

A bound sampling profile specifies the signal representation, sample clock, sample spacing, any kernel/filter, normalization and relationship to the batch interval. Its output adapter supplies the typed increments consumed by the chart lane. Batch-label equality remains distinct from measured inter-channel phase coherence.

19.1 Three-dimensional input requires a declared map

A 3D-to-2D adapter must identify its projection, slice or other reduction and what information it discards. A linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$ cannot be injective: its rank is at most two, so its nullspace has at least one dimension. The master cannot silently claim lossless general 3D data in a two-scalar record. [D, line 141]

The reference fixture uses `identity_2d`. It does not implement a 3D reduction under a misleading two-dimensional name.

19.2 OTAN2 is not an absolute-arrival estimator

D calls the final readout a localized direction-of-arrival orientation. The preserved formula takes only a chart-increment pair and an axis. Two observations with different absolute phase origins but identical $(\Delta\rho, \Delta\phi, \alpha)$ necessarily produce the same OTAN2 result. Therefore that formula alone cannot determine both distinct absolute origins or reconstruct a raw incoming-wave bearing.

The master retains D's intended connection as a *proposed upstream-to-chart pipeline*, but names the computed result exactly: log-increment orientation relative to a supplied axis. An actual direction-of-arrival estimator would require its own calibrated observation map and input model. This explicit distinction does not alter the source's literal OTAN2 equation. [O, pp. 6–7][D, lines 35, 96][N06]

IV / OTAN2 AND THE INVARIANT STATE

20 The literal OTAN2 operator

The author explicitly distinguishes *otan2* from ordinary *atan2*. The final source definition is retained as the default:

$$\text{otan2}_s(d_\phi, d_\rho) = \arctan\left(\frac{d_\phi}{d_\rho}\right), \quad d_\rho \neq 0, \quad (40)$$

$$e_s = \text{otan2}_s(d_\phi, d_\rho) - \alpha. \quad (41)$$

Here $d_\rho = \Delta\rho$ and $d_\phi = \Delta\phi = \Delta\theta$ in the selected source alias. The reference-axis orientation α must use the same log-tangent frame. [O, pp. 5–7]

20.1 What the source supplies

The static chart separates logarithmic radial change from angular change. The pinion/spoke language describes a local changing vector. The moving axis supplies a relative reference. The subtraction does not change the chart or rewrite its LUT.

The ratio is undefined when $d_\rho = 0$. When both increments are zero there is no direction of change, independently of the ratio problem. The source's arctangent range is $(-\pi/2, \pi/2)$ on its finite ratio domain. No quadrant-restoration bit or final wrap appears in the literal final source formula.

20.2 Default executable tag

The code tag is `source_ratio`. The raw residual is stored exactly as the source's subtraction, subject to the finite numeric representation. Missing or singular inputs are statuses, not silently assigned numerical zero.

The earlier source material also defines direct phase projection and wrapped phase difference. Those are retained as separate helpers. A phase of a point, a difference between phases and an orientation of an increment pair have different input and output meaning.

20.3 Literal reversal ambiguity

For nonzero denominator, simultaneous reversal leaves the ratio unchanged:

$$\text{otan2}_s(-d_\phi, -d_\rho) = \text{otan2}_s(d_\phi, d_\rho). \quad (42)$$

This is an exact property of the supplied expression. The master does not hide it by claiming the ratio already retains a complete directed quadrant. It instead preserves the previously named optional completion as a separate profile.

21 Directed completion and angle views

The integrated family retains `directed_completion_1` as an explicit alternative:

$$\text{otan2}_d(d_\phi, d_\rho) = \text{wrap}_{2\pi}(\text{atan2}(d_\phi, d_\rho)), \quad (d_\rho, d_\phi) \neq (0, 0), \quad (43)$$

$$e_d = \text{wrap}_{2\pi}(\text{otan2}_d - \alpha). \quad (44)$$

This is a retained completion decision from the prior master family. It is not a relabelling of O's one-argument ratio. The selected profile is recorded in the output.

For period $p > 0$, use

$$\text{wrap}_p(u) = u - p \left\lfloor \frac{u + p/2}{p} \right\rfloor \in [-p/2, p/2). \quad (45)$$

The result can carry three distinct views:

$$e_{\text{raw}} = \beta - \alpha, \quad e_{\text{principal}} = \text{wrap}_{2\pi}(e_{\text{raw}}), \quad e_{\text{line}} = \text{wrap}_\pi(e_{\text{raw}}). \quad (46)$$

The line-class view does not overwrite the raw source residual or select an unprovided eigenvector-sign convention.

d_ρ	d_ϕ	Source ratio	Completion
1	1	$\pi/4$	$\pi/4$
-1	1	$-\pi/4$	$3\pi/4$
-1	-1	$\pi/4$	$-3\pi/4$
0	1	Undefined ratio	$\pi/2$
0	0	Zero increment	Zero increment

21.1 Winding and interval policy

The main evaluator accepts explicit accumulated increments. The endpoint helper requires either `known_turns` with an integer or a named `principal_difference` convention. It never infers a missing turn count from two wrapped samples.

A common positive interval cancels in the ratio, so rates and increments yield the same orientation only when they share that interval and units. The coherent-batch layer records the interval rather than deriving a duration from an author date or from a byte offset. [D, lines 94, 142]

22 Definition statuses are part of the operator

Status	Meaning in the reference
defined	Required domain and representation conditions hold.
zero_increment	Both components are zero; no orientation assigned.
ratio_undefined	Nonzero phase increment and zero literal denominator.
nonfinite_input	A required numerical input is not finite.
numerical_range	Intermediate arithmetic leaves the declared numeric representation.
axis_unspecified	Orientation may exist but the relative reference is absent.
frame_unspecified	The expected common log-tangent frame is absent.
increment_policy_unspecified	The required explicit-increment policy was not selected.
turn_information_unspecified	The endpoint helper lacks its declared lift information.
profile_unspecified	Neither supported orientation profile was selected.
input_unspecified	A word record requested no orientation input.

The orientation status and relative-readout status are independent fields. With an absent axis the reference can retain a valid β and leave residual fields null. This is more informative than setting every value to zero.

22.1 Numerical range and local test guards

A literal quotient that overflows, or underflows a nonzero numerator to zero, is reported as `numerical_range`. The optional frame probe additionally detects a rotated literal denominator within $64\epsilon\|v\|$ of zero and records that comparison as unavailable. This numerical guard does not change the base equation's exact mathematical domain.

The invariant tolerance is 10^{-10} radians. It belongs to the finite comparison routine, not to a physical alignment threshold, a beamwidth or a J/K feedback policy. An angular residual outside an application's desired range is still an ordinary defined value until a separate policy defines an interpretation.

22.2 Required-input error versus optional undefined result

Malformed mandatory batch input rejects the epoch before publication. A finite input at an OTAN2 singularity produces a valid diagnostic record while the independently defined core and JK result may commit. The implementation includes regression tests for both paths, including a valid run immediately after a rejected batch.

23 The invariant bank within the engine

The diagnostic state contains four preserved-value identities and two profile laws. Each entry records its domain, transformation, comparison space and observed error.

ID	Transformation	Expected relation
R	Common positive radial scaling	The common log-radius offset cancels in the increment.
W	Common positive interval	Dividing both components by the same W preserves orientation.
P	Common phase-origin shift	The difference of consistently lifted phases is unchanged.
S	Positive common step factor	The increment ratio and directed orientation are unchanged.
F	Joint frame rotation	Co-rotated increment and axis preserve the source line class or directed class.
V	Reverse both components	Source ratio is unchanged; directed orientation changes by a half-turn.

[O, pp. 6–7][M1, invariant bank]

23.1 R, W and P

For $c > 0$, $\rho_i \mapsto \rho_i + \ln c$ gives the same $\rho_1 - \rho_0$. For $W > 0$, $(d_\phi/W)/(d_\rho/W) = d_\phi/d_\rho$. Adding a common γ to both lifted phases likewise cancels in their difference. Each identity requires its own stated common reference, interval or lift.

Changing a physical origin, independently resampling two channels or wrapping their endpoints under different policies is not one of those transformations. The invariant registry does not silently expand its claims to include them.

23.2 S and the source spiral interpretation

For $c > 0$,

$$\frac{cd_\phi}{cd_\rho} = \frac{d_\phi}{d_\rho}. \quad (47)$$

For a supplied chart line $\phi = k\rho + b$, every nonzero radial segment yields $\text{atan2}_s = \arctan k$. This is the constant-pitch example in the source, with an explicit domain. If the reference axis moves, the relative residual responds to that motion.

A **PASS** means the declared numerical relation was satisfied for the tested snapshot. A **FAIL** means the completed comparison exceeded tolerance; **UNDEFINED** means a required comparison value or precondition was unavailable. No entry becomes a use-case authorization decision.

24 Frame and reversal proofs

Let $v = (d_\rho, d_\phi)^T \neq 0$ and let R_χ rotate vectors in the declared log-tangent frame. Rotate the reference at the same time: $\alpha' = \alpha + \chi$.

Theorem 24.1 (Joint frame relation). *The directed relative angle is unchanged modulo 2π . The literal relative angle is unchanged modulo π , where both source ratios are defined.*

Proof. The directed vector argument increases by χ modulo 2π . The reference subtraction removes the same χ . The literal ratio represents only an unoriented slope, because simultaneous sign reversal leaves it unchanged. Its cancellation therefore holds in the line class modulo π . $\square$

24.1 Raw values need not agree

Start with $v = (1, 0)$ and $\alpha = 0$. Rotate both by $3\pi/4$. The rotated source angle is $-\pi/4$ and the new raw residual is $-\pi$. The old raw residual was zero. Their line classes agree, while the completion's principal residual agrees directly. The F comparison uses the appropriate space instead of hiding this branch behavior.

Proposition 24.2 (Reversal relation). *For a defined source ratio, $\text{otan2}_s(-v) = \text{otan2}_s(v)$. For the directed profile, $\text{otan2}_d(-v) = \text{otan2}_d(v) + \pi$ modulo 2π .*

Proof. The quotient is unchanged under two sign reversals. A nonzero directed vector points oppositely after reversal, changing its argument by one half-turn. $\square$

24.2 Axis-only movement is not an invariant

Keeping v fixed while changing α by η gives

$$e'_s = e_s - \eta, \quad e'_d = \text{wrap}_{2\pi}(e_d - \eta). \quad (48)$$

This is the intended covariance of the moving reference. A panel that never changed when only its reference moved would not be displaying the source's subtraction. The master stores this observation alongside the invariants without calling it a failed source identity.

V / THE FULLY COMPOSED EPOCH

25 The complete execution sequence

The master now places the source roles, extension declarations and reference operators in one execution contract. Its selected profile is explicit; no new field, encoder or policy is inferred at runtime from prose.

Stage	Operation	Dependency and result
0	Resolve selected profile	Record version, dictionary, geometry, field/transfer bindings and policy identity.
1	Validate coherent batch	Check required fields, epoch, snapshot identity, clock/frame labels, half-open times, word completeness and event ordering.
2	Freeze input snapshot	Copy configuration and input records; keep their content identity.
3	Resolve typed observations	Determine alignment/limit/fringe record statuses and explicit encodings. Any index conversion uses the selected transfer.
4	Evaluate core	Apply ASA, NA, population count and whole-word disposition in canonical logical groups.
5	Apply JK events	Update the stored bit sequentially from supplied pairs. No angular assignment is inferred.
6	Evaluate OTAN2/bank	Compute requested orientation/status views and six comparisons from the declared increments and axis.
7	Assemble epoch record	Include old/new JK values, all intermediate words, diagnostics, interval and source/profile identities.
8	Commit	Append the hash-linked event and publish next state after all required work succeeds.

[D, lines 49–50][M1, master epoch][N07]

Stages without dependencies can be computed in a different order provided the result is equivalent to this frozen-snapshot contract. The current Python reference is synchronous; it does not claim a concurrent transactional store or GPU scheduler.

A malformed required field yields an error and the old state. A missing optional orientation value or a source-ratio singularity yields a diagnostic status. The boundary mask’s zeroing rule remains computational disposition, not a remote disable operation.

Source “precedence 0” is retained as dependency/interpretation order. Neither D nor the earlier source supplies a zero-duration execution guarantee, kernel-mode privilege transition or device dispatch protocol through that label.

26 Atomicity, auditability and ordinary errors

For a valid local batch, the next implemented state is

$$S_{\text{local}}^+ = \langle n + 1, Q^+, (y_i)_i, t_1, h^+ \rangle. \quad (49)$$

Before that assignment, the reference computes the entire result and canonical hash-chain input. Required failures publish none of those next-state fields.

Proposition 26.1 (No partial epoch after a validation failure). *A rejected batch leaves the reference's state and committed-event list unchanged.*

Proof. All required validation and result construction occur on copied local data. The only publication occurs after those stages and after construction of the next state and event. A validation exception occurs before publication, so the old fields and list remain unchanged. $\square$

This is an in-process synchronous property, not a claim that a disk write is automatically crash-atomic. The CLI can write the completed result to a new file; durable transactional storage would need its own implementation contract.

26.1 A rejected input does not lock the engine

The tests mutate a valid batch into mixed-clock, stale-snapshot, invalid-tail and other malformed variants. After each rejection they submit the original valid batch and verify normal progress. There is no persistent refusal state, remote authorization request or hidden use-case classification in that path.

The batch/domain rules are retained because D expressly requires coherent data and resolved types. Removing them while still claiming the same coherent-batch formalization would change the model, not merely remove a banner. [D, lines 45–50, 137–142]

26.2 Data remains data

JSON parsing does not evaluate code strings. Duplicate keys are rejected rather than allowing a hidden later value to replace an earlier field silently. A source quote, author token or filename is not invoked as a command. Those are ordinary representation rules for reproducible records.

The master contains no mechanism to disable platform protections. No driver or hardware interlock is modified by the supplied local package. Its runtime dependency is ordinary Python arithmetic and files explicitly selected by the user.

27 Integrated records and deterministic replay

A committed epoch contains

$$R_n = \langle \text{profile}, \text{batch}, \text{clock}, [t_0, t_1), \text{words}, \text{JK}, \text{blend}, \text{OTAN2}, \text{bank} \rangle. \quad (50)$$

Each word result stores (a, n, h, y) , predicate records and any calibrated orientation inputs. JK stores its pre-state, post-state and ordered event trace. No successful diagnostic is substituted for another subsystem's result.

27.1 Concrete reference schema

```
{
  "schema": "aTOMos-epoch-3.6.1",
  "engine": "aTOMos", "version": "3.6.1",
  "profile_id": "local-mathematical-reference/1",
  "profile_sha256": "...", "epoch": 0,
  "batch_id": "fixture-batch-0000",
  "clock_id": "fixture-clock",
  "start_ns": 0, "end_ns": 125000000,
  "records": [
    { "word": 0,
      "core": { "asa": 15, "na": 14, "hits": 0, "output": 14 },
      "blend": { "status": "defined", "value": 0 },
      "otan2": { "profile": "source_ratio", "status": "defined" },
      "invariants": { "R": {}, "W": {}, "P": {},
                     "S": {}, "F": {}, "V": {} }
    },
    { "jk": { "before": 0, "after": 1, "events": [] }
  ]
}
```

This abbreviated illustration omits other word records and detailed events; the complete actual export is in `examples/expected_epoch.json`.

27.2 Replay and input-order identity

A fresh engine with the same profile and batch emits the same complete record under the same numerical semantics. Changing the input record order leaves canonically sorted word outputs unchanged, but changes the input digest because the received list order is evidence too.

Committed state exposes immutable tuples for its words. Returned profiles and event lists are copies. Editing a returned report does not change the stored history. These properties are tested directly rather than assumed from a hash label.

28 Provenance and canonical serialization

The reference uses a tagged canonical encoding before hashing. Null, Boolean, integer, floating, string, list and map types have distinct tags. Map keys are sorted by UTF-8 bytes. Integer values use decimal text; finite floating values use binary64 hexadecimal text. Numeric -0.0 is normalized to $+0.0$.

This avoids a collision between a real floating value and a user dictionary merely shaped to look like a float wrapper. It also avoids claiming that the integer 1 and float 1.0 are the same source record. Type preservation is a deliberate new serialization profile. [N08]

A profile digest is

$$h_{\Theta} = \text{SHA256}(d_{\Theta} \parallel \text{len64}_{BE}(C(\Theta)) \parallel C(\Theta)). \quad (51)$$

An epoch chain uses

$$h_{n+1} = \text{SHA256}(d_E \parallel h_n \parallel \text{len64}_{BE}(C(R_n)) \parallel C(R_n)). \quad (52)$$

The initial chain head in the reference is the profile digest. Distinct domain strings identify record, profile, genesis and epoch roles. The full values are retained rather than squeezed into a one-bit identity field.

28.1 What the verifier establishes

Given a retained initial head and event list, the verifier recomputes each link and checks predecessor equality. Altering a result word without updating its linked record invalidates the check. Replacing an entire unanchored ledger and all its hashes can remain internally consistent; this mechanism is an integrity reference, not a signature or proof that an input was measured.

28.2 Future versions preserve derivation history

Earlier engine editions used different canonical forms and domain strings. S1 does not silently recalculate their identities and call the results unchanged. A migrated record names its original serializer/profile and any new derivation relationship.

The package-level SHA-256 manifest separately covers delivered files. It is verified without network access. Editing an artifact intentionally changes its byte digest; the old manifest then documents the prior bytes rather than blocking the user's edit.

29 Compatibility of the upgraded master

Theorem 29.1 (Inherited word projection). *In record mode, projection of a v3.6.1 epoch onto each four-word ASA result equals the inherited v3.6 calculation on the same words and masks.*

Proof. The effective fringe is the valid mask. All four stages therefore receive the same operands as the inherited rule. The additional batch, source and diagnostic records do not alter those operands. $\square$

Theorem 29.2 (Explicit JK and diagnostic isolation). *When the selected JK policy is the supplied event list, changing only the reference axis or an optional OTAN2 definition status does not change the JK transition or core word outputs.*

Proof. JK reads only its pre-state and explicit J/K events. The word core reads occupancy and masks. Neither function reads the orientation result in this profile. Their projections are therefore independent of that changed diagnostic input. $\square$

29.1 Neutral interface extension

Adding an unbound physical-profile declaration to the foundational registry changes the profile document but does not provide numerical values to a local word epoch. Selecting the fully bound local reference gives a complete calculation from its own specified inputs. The two profiles are not falsely identified as the same deployed hardware configuration.

29.2 Behavioral variants need explicit maps

A future feedback policy which consumes OTAN2 and changes J/K is a new behavioral variant. Its disabled or neutral mode may recover the inherited projection, but that recovery needs its own statement. The current source does not supply such a map, so the reference does not invent one under the phrase “filled profile.” [D, lines 37–43, 133]

29.3 State-machine totality

The local evaluator is defined on its typed coherent-batch domain. Source singularities in optional diagnostics are represented within the result type. Malformed mandatory input is outside that domain and leaves the old state. Calling the engine total does not require silently assigning a numerical angle to an undefined mathematical case.

VI / LAYOUT, PROFILES AND VERIFICATION

30 Word packing, Morton layout and the texture contract

For R radial rows and P angular cells, define $W = \lceil P/32 \rceil$. Each uint32 word stores 32 logical angular bits, with the final word's unused high bits zero. The canonical result order is $q = iW + w$.

For the inherited 8-by-8 word-tile profile, pad rows and word columns to multiples of eight. Within a tile,

$$z(i, w) = \sum_{k=0}^2 \left(\text{bit}_k(i)2^{2k} + \text{bit}_k(w)2^{2k+1} \right). \quad (53)$$

Radial bits occupy even positions; angular-word bits occupy odd positions. Tile order is row-major. This preserves the source's interleaving direction while making its word granularity explicit. [A, pp. 85–87][M1, storage profile]

Theorem 30.1 (Storage permutation equivalence). *If each input plane stores its logical word (i, w) at the selected bijective address $\pi(i, w)$, linear and Morton retrieval followed by the same word rule produce identical canonical output words.*

Proof. The tile quotient and remainder are unique. The even and odd address bits recover the two within-tile coordinates, proving invertibility. Retrieving address $\pi(i, w)$ thus returns the same logical operand from every plane; the arithmetic and canonical output index are identical. $\square$

30.1 The local code and GPU boundary

The reference includes the forward/inverse address functions and tests non-multiple padded dimensions. It performs no GPU allocation. A future texture backend must preserve the same logical inputs, immutable snapshot and independent output storage.

Texture resources contain data; native instructions execute separately. The prior project target remains the user-requested RTX 5070 Ti Laptop 12 GB profile, recorded as deployment context, not a newly observed device or benchmark. This package contains no compiled CUDA artifact and no custom host ring-0 driver.

The source's final swizzle is a layout concept, not a semantic filter or a guarantee that all needed words are permanently resident in a cache. A performance claim belongs to an actual backend run with its device, compiler, workload and measurement interval recorded.

31 Dimensions, storage cost and explicit resources

A logical word contains 32 bits regardless of whether its predicate describes one region or another. An identifier and a sampled real quantity retain their own representation. The entire engine is therefore not a one-bit allocation even when its geometric predicates are packed.

For four input planes with padded size $S = R_p W_p$ and $L = RW$ four-word outputs,

$$M_{\text{payload}} = 16S + 16L \text{ bytes.} \quad (54)$$

A separately stored fringe plane adds $4S$. Hashes, matrices, states, metadata and event records add further memory. The Python reference uses ordinary high-level objects, so this binary-layout formula is not its actual heap usage.

31.1 Finite workloads

The word arithmetic has constant work per logical word and $O(L)$ total work for L words. The source grammar can double the number of proposals per generation; without pruning, the depth- g frontier has 2^g members. Constant-time dictionary lookup does not make that complete output constant-size or constant-time.

A numerical producer adds its declared number of field evaluations, and the invariant bank adds finite comparison probes. Serialization requires work proportional to emitted record bytes. Source performance superlatives are not imported as benchmarks of this implementation.

31.2 No remote or semantic authorization layer

The local engine does not consult a license service, an author-identity allowlist or a use-case classifier. Its resource/domain checks are explicit code and schema constraints. A rejected integer range or a duplicate word is a representation error, not a restriction inferred from the content of a source label.

An actual accelerated deployment should declare allocation plans and handle ordinary allocation failures. Removing a mathematical bound while continuing to claim the same finite layout would change the representation contract. The package does not modify operating-system settings or device interlocks to avoid such failures.

32 The optional programmable computation profile

The family retains its separate universal-interpreter profile. It is not a theorem that the fixed ASA word filter or OTAN2 expression alone simulates every program.

Let Q be a finite control set, H a halt set and $T : \mathbb{Z} \rightarrow \{0, 1\}$ a tape with finite nonzero support at each finite step. A program supplies

$$\delta : (Q \setminus H) \times \{0, 1\} \rightarrow Q \times \{0, 1\} \times \{-1, 0, 1\}. \quad (55)$$

If $\delta(q, T(p)) = (q', b, d)$, one transition is

$$(q, p, T) \mapsto (q', p + d, T[p \mapsto b]). \quad (56)$$

[M1, optional U profile]

Theorem 32.1 (Encoded transition simulation). *With a matching initial configuration and adequate logical storage, a fixed interpreter of the table preserves every defined transition of an encoded deterministic binary-tape machine and its first entry into a halt state.*

Proof. Initially copy the control, head and finite input. If the representations agree before a step, the interpreter reads the same symbol, selects the same rule, writes the same cell, shifts the head by the same amount and selects the same state. Induction proves agreement for each defined finite prefix and therefore preserves halt-state entry. $\square$

The supplied TapeMachine utility uses a sparse logical tape with integer addresses. It distinguishes halted, missing_rule, running and budget_exhausted. Exhausting a finite execution budget is not silently called a mathematical halt.

This profile shares the family attribution and can use the same record/provenance conventions. It does not share the word filter's update equation or require a spatial bitmask for its simulated tape. The implementation is a local interpreter utility, not privileged execution of text supplied inside source documents.

33 The completely populated local reference profile

The purpose of the worked profile is to close the local arithmetic and integration example without pretending that D supplied actual hardware constants. Its configuration is explicit:

Field	Bound reference value
Profile ID	local-mathematical-reference/1.
Dictionary	Two rows, 40 angular cells; four logical uint32 words, logical row-major order.
Masks	Explicit ASA, NA, boundary and fringe word arrays in the JSON profile.
Input model	local_preprocessed_words/1.
OTAN2	source_ratio; explicit log-tangent frame.
Index conversion	$h_\rho = 0.125$, $h_\phi = 0.0625$ radians per index.
B test field	Constant (0.25, 0.5) in the declared numerical test units.
Matrix fixture	diag(2, 1), selected vector (1, 0) and eigenvalue 2.
Dimensional adapter	identity_2d; no unrecorded 3D projection.
Sampling representation	Provided local records; 125,000,000 ns example interval and synthetic clock label.
JK input	Explicit ordered event pairs; initial $Q = 0$.
Raw hardware binding	Not bound or executed in this profile.

33.1 What “filled” means here

Every field needed by this implemented profile has a value or an explicit not_applicable/unbound role. The test field and matrix are not copies of a missing antenna calibration. Their provenance is authored_test_fixture. The foundational profile separately records exactly which entries remain absent from D for a physically bound variant.

The equation tests include the constant-field RK4 result and matrix identity, while the word epoch uses its own preprocessed inputs. A test utility is not silently run as a physical propagation stage merely because it appears in the same profile document.

The profile itself is part of the input snapshot digest. Editing a mask, interval interpretation, transfer factor or selected profile creates a new snapshot to which a new batch must refer.

34 A complete worked epoch

The example batch uses occupancy words

$$X = [15, 165, 257, 85], \quad (57)$$

with ASA masks $[15, 255, 511, 255]$, NA masks $[14, 255, 511, 255]$ and boundary masks $[0, 128, 256, 0]$. Record-mode fringe is neutral. Each row's second word has only eight valid bits because the angular count is 40.

Word	x	a	n	h	y
0	15	15	14	0	14
1	165	165	165	1	0
2	257	257	257	1	0
3	85	85	85	0	85

An explicit set event changes the initial JK state from zero to one. The supplied blend bits $(1, 0, 1)$ produce zero parity for each example record. Alignment, sweep-limit and active-fringe classifier records remain unspecified because the fixture does not pretend that encoded blend parity supplies all three meanings.

Each orientation input uses $(\Delta i, \Delta j) = (4, 8)$ with the declared scales, producing $(\Delta \rho, \Delta \phi) = (0.5, 0.5)$ and axis $\alpha = 0$. Therefore

$$\beta = e_{\text{raw}} = \pi/4. \quad (58)$$

All six comparisons satisfy the selected identity/profile law on this sample. The full result includes each error value and comparison space rather than a single unqualified “valid” flag.

34.1 Two contrasting mutations

Changing the first orientation's radial index change to zero makes its literal OTAN2 status `ratio_undefined`. The first core output remains 14 and the explicit JK event still sets the bit. Changing the first record's clock ID instead makes the coherent batch invalid before any state publication.

These cases distinguish an unavailable mathematical observation from a malformed mandatory input. The blue-team suite verifies both the output separation and the absence of a persistent lockout after a rejected batch.

35 The foundational profile as a future-variant substrate

D identifies three extension entry points: the snapshot/envelope, the input producer and a future JK policy. It also identifies the field, matrix and transfer obligations. S1 joins them as one schema family rather than leaving separate informal append prompts. [D, lines 23–50, 74–96, 124–148]

A variant record has the conceptual form

$$V = \langle \text{id}, \text{parent digest}, \text{replacement fields}, \text{sources}, \text{domain}, \text{tests} \rangle. \quad (59)$$

The parent is content-addressed. Each replacement names its previous role, its new value and its provenance class. A field may be supplied from a source, mathematical derivation, measurement or authored fixture; the variant does not collapse those evidence classes.

35.1 Stable base relationships

Future variants retain the base names and type boundaries unless explicitly versioning them. ASA-after-input and NA-after-ASA retain their intermediate records; the source OTAN2 remains identifiable; a new feedback profile must identify its J/K mapping. A schema change that alters a word's cell grouping or numeric frame is not described as a harmless renaming.

35.2 A declarative registry is not execution authority

The reference hashes and stores profile data. It does not execute a string containing a differential equation, assembly fragment or import statement. A code implementation of a new field or adapter is reviewed as code and selected through an explicit interface; its prose description alone is not executable.

No centralized registry is required by S1. The author can maintain local branches and issue new profiles, with the old profile/digest retained for replay. This satisfies the requested foundational role without introducing a remote control plane.

35.3 Completion accounting

A physically bound profile can mark an obligation complete only when it includes the actual content, units/frame and tests relevant to that obligation. A test-field fixture demonstrates the interface; it does not automatically complete an unrelated material or array profile. This is a record of supplied knowledge, not an automated decision about the eventual application.

36 Source claims, formal definitions and explicit clarifications

This release follows the attachment's organization while keeping its unsupported equivalences visible. The following table is part of the integration, not an unannounced rewrite.

Source statement or role	Master treatment
D says the engine is a localized spatial aperture processor.	Retained as the name and role of the preprocessed-input profile. The exact word processing is fully stated.
D says physics can append to RK4's field.	Retained as a field-binding obligation; actual equations, dimensions and boundary/initial values are not supplied.
D uses B-profile for medium properties.	Kept distinct from B's earlier kinematic descriptor. A variant may explicitly relate them.
D calls an eigenmatrix a set of beamforming coefficients.	Separate axis-reference and aperture/observation-matrix roles. Their equivalence is not implied by the label.
D equates index scaling with an optical/phase transfer.	The supplied scaling equation is implemented for preprocessed indices; the unprovided analog transfer remains separate.
D calls OTAN2 a final direction-of-arrival orientation.	Retain the intended upstream link but distinguish the actual increment-orientation formula from an absolute raw-wave estimator.
D prohibits mixing clock frames.	Enforce record coherence and interval membership; do not claim a clock label proves physical phase calibration.
D cites specific sections of pdf_v9BtD7.pdf.	Retain those as source citations. The opaque file is not separately inspected; this master provides its own anchors and source-line map.

The original attachment is preserved exactly in the ZIP so every clarification can be compared with its source. Its mathematical slots are incorporated throughout the engine rather than returned as a disconnected essay. No missing parameter is silently replaced with a convincing-looking constant.

37 Blue-team verification plan

The supplied tests exercise the local data and arithmetic contracts with both valid and adversarially malformed fixtures. They do not require network access, external accounts or an AI service.

Test family	Required observation
Integer domains	Reject nonwords, out-of-range values and Boolean/integer confusion.
Core semantics	Compare all four fields with a separate per-bit oracle; preserve whole-word absorption.
State updates	Verify the entire JK table, explicit event order and unchanged state after rejection.
Coherent input	Detect mixed batch/clock/frame/profile IDs, duplicate/missing logical words and out-of-interval times.
Temporal replay	Reject repeated live epoch without changing committed state; replay fresh state deterministically.
OTAN2	Preserve source/completion difference, zero/singularity statuses, phase policy and axis covariance.
Invariant bank	Test all four invariants and two profile laws under bounded random transformations.
Serialization	Reject duplicate JSON keys and nonfinite canonical values; distinguish scalar and forged-wrapper types.
Provenance	Detect a changed epoch under a retained head and preserve exact predecessor linkage.
Recovery behavior	A rejected malformed batch must not prevent the subsequent original valid batch from running.

37.1 Independent oracles

The Python unit suite's per-bit word oracle uses sets of bit positions rather than the word expression under test. An additional JavaScript oracle implements independent bit selection and orientation equations, then compares complete values against Python over deterministic input sets.

The tests also cover the standalone RK4 constant-field and refinement cases, Morton inverse maps on padded dimensions, lineage IDs, symbolic genesis normalization and the separate tape interpreter's halt/budget statuses. Those utilities are part of the full architecture, while the CLI's executed producer remains local preprocessed words.

A failing test is a reproducible engineering finding, not a license or application denial. The tests and implementation are included for inspection and revision.

38 Executed evidence for v3.6.1

The current package was run in the preparation environment. Counts below are from this release's files, not copied from earlier chat claims.

Executed check	Result
Python unit/regression suite	68 tests; zero failures, errors or skips.
Independent JS/Python words	2,048 word cases, all four result fields compared.
Independent JS/Python angles	2,018 cases; profile/status and defined values compared.
Cross-runtime total	14,238 comparisons passed.
Python runtime	3.13.5 in the preparation run.
Node runtime	v22.16.0 for the independent oracle.
Golden CLI output	Matched the complete saved local epoch.

38.1 What the Python suite executed

The 68 unit tests include loops over every 16-bit population-count input, 3,000 deterministic random word tuples, bounded random invariant cases, full small truth tables and malformed coherent batches. A unit-test count is not the same as the number of scalar comparisons inside its loops; the report therefore does not inflate those into an unspecified global assertion total.

38.2 Independent JavaScript comparison

The optional cross-runtime check evaluated 2,048 word cases and 2,018 angle cases. It compared 14,238 scalar/status values with the Python reference. The maximum observed absolute angular difference was $8.881784197001252 \times 10^{-16}$ radians, below the declared 10^{-12} comparison tolerance.

38.3 Concrete implementation coverage

The local word epoch, coherent input parser, profile binding, explicit JK events, blend, OTAN2, invariant bank and hash record pipeline were executed. RK4, layout, lineage and tape utilities were exercised by their specific tests. A raw array acquisition stack, a physical propagation model, a hardware feedback loop and CUDA execution were not run or claimed by those results.

The complete mathematical master includes the interface slots for those variants. The new attachment supplies their names and relationships, not their numerical implementations. This evidence boundary is recorded in machine-readable form at `results/validation_status.json` and in the source/implementation matrix, rather than hidden in a generic success label.

39 Implementation and profile conformance matrix

Master area	Delivered artifact	Conformance scope
State and master word epoch	atomos/engine.py	Synchronous local epoch, canonical word order and no partial publication.
ASA, blend, JK and word grammar	atomos/core.py	Exact finite-word functions and truth tables.
OTAN2 and invariants	atomos/orientation.py	Source/completion/status views and six checks.
Chart, layout and RK4	atomos/geometry.py	Pure utilities with declared inputs and tests.
Identity and replay record	atomos/records.py	Canonical tagged representation, genesis and event-chain verifier.
Universal profile	atomos/tape.py	Sparse local interpreter, finite execution budget and explicit statuses.
Foundational substrate	profiles/foundation.json	All D slots, source locations, required bindings and variant rules.
Complete local profile	profiles/local_reference.json	Authored test inputs, not hardware calibration.
Schemas	schemas/*.json	Structural record declarations; cross-record constraints also enforced in code.
Physical field and arrays	Declared profile slots	Actual numeric model, coefficients and calibration not supplied by D.
GPU texture backend	Master layout/interface contract	No new CUDA binary or GPU execution in this ZIP.

A source-code review should follow the table rather than assume that inclusion of a formula in the PDF means every possible backend was compiled. The complete engine architecture and the delivered reference implementation have explicitly different coverage scopes.

All executable modules are local files. There is no deferred online action, hidden plugin, credential lookup or unattended background task. The package can be inspected without invoking a model or network service.

40 Reproduce, inspect and extend

From the extracted directory, run:

```
python -m atomos --profile profiles/local_reference.json \  
  --batch examples/coherent_batch.json  
python tools/run_tests.py  
python tools/crosscheck.py
```

The first command emits the complete local epoch JSON. The second runs the Python suite. The third runs the optional Node/Python comparison. Python 3.10 or later is sufficient for the runtime and primary tests; Node is needed only for the independent JavaScript oracle.

40.1 Editable source and document build

The PDF's editable $\text{\LaTeX}$ source and source maps are included. To rebuild the fixtures and PDF:

```
python tools/make_examples.py  
sh tools/build_pdf.sh
```

The document build needs a $\text{\LaTeX}$ distribution with the packages named in its preamble. It is not a runtime dependency of the engine reference.

40.2 Review task

Read `AGENTS.md`, `docs/CONTRACT.md` and both profiles. Reproduce a failure before repairing it, add a regression test and record what changed. Keep the source-ratio profile, angle classes, predicate meanings and neutral-mask identity visible. Do not weaken an oracle or replace an undefined result with zero to obtain a pass.

An actual scientific or hardware variant supplies its field, calibration and observation/control bindings with units and tests. It can then select that named implementation in a separate version. The local reference is the concrete baseline against which unchanged word and record semantics can be compared.

The integrity command `python tools/verify_package.py` checks the unsigned delivery manifest. Editing a file is allowed; the old hash then accurately identifies an earlier version rather than preventing the edit.

41 Current consolidation decisions

ID	Decision	Effect
N01	Rebuilt master and source mapping	One v3.6.1 master, M2, from the established core and available sources. Opaque source references are not silently verified.
N02	S1 profile registry	Source obligations and actual bindings are separate; the local fixture is fully populated for its own declared model.
N03	Namespaced physics/matrix roles	Original B-kinematic and proposed B-medium roles remain distinguishable; axis matrix is not automatically an aperture matrix.
N04	Evolution-compatible field port	RK4 requires a declared first-order state/field; named wave physics is not automatically a two-coordinate derivative.
N05	Concrete coherent batches	Versioned snapshot, clock, frame, half-open integer times, word uniqueness and predicate status before selection.
N06	OTAN2 meaning preserved	Source ratio remains default; absolute raw-wave arrival estimation is a separate unprovided adapter, not inferred from the ratio.
N07	Integrated atomic epoch	Compute the source/core/JK/diagnostic record over a frozen batch and publish only a complete result. Undefined optional diagnostics do not implicitly reset other layers.
N08	Tagged canonical provenance	Preserve value types, finite float representations, key ordering and domain-specific digests. Historical encodings are not rewritten as identical.
N09	Transparent local package	No remote activation, license gate, telemetry or use-case classifier. Mathematical/data-domain checks are ordinary inspectable code.
N10	Actual validation record	New local tests, independent oracles, golden examples and a stated coverage matrix; no unrun physical/GPU success claimed.

These are explicit formalization decisions. They are not presented as text already authored inside the attachment. The source's own terminology, order and open questions remain visible in the source map and preserved original file.

The foundational role is a stable contract for future variants, not a statement that every future medium, sensor or policy has already been numerically specified. A future binding becomes part of the same architecture through a versioned, testable profile.

42 Source register

Key	Source	Extent
A	TOMKLOOTWIJKWHITEKINGballistictrajectorycloudchamber(1).pdf; original core, identity, JK, ASA/NA and layout material.	91 pages
F	TOMKLOOTWIJKfly's-have-interesting-eyes-how-do-they-work(1).pdf; faceted-input and edge/central architecture.	8 pages
B	TOMKLOOTWIJKTHEblendoperatorterrorpredatorjellyjiggledrone.pdf; A-W-B-N, predicate meanings and hollow/fringe roles.	5 pages
O	TOMKLOOTWIJKORANGENATOotan2operatormath(1).pdf; literal ratio and moving-axis source.	7 pages
D	Pastedmarkdown.md; current configuration, coherent-batch, physics-slot and future-policy proposal.	152 lines
M1	Previous atomOS v3.6 total integrated master contract established in this conversation. Used as inherited definitions; its PDF file was not mounted in this run.	Prior chat

The original source pages were inspected through their extracted text and relevant rendered pages. O’s final equations on pp. 6–7 are central to the preserved OTAN2 definition. D’s complete 152-line content is retained in the ZIP with its exact byte digest. The older private PDFs are fingerprinted rather than redistributed.

D repeatedly refers to pdf_v9BtD7.pdf; that handle alone does not provide its bytes or certify its reported section anchors. This master cites D’s own line ranges and gives a fresh, complete integrated specification instead of inventing a resolved file behind that handle.

No outside physics model or calibration research is inserted into the source narrative. The new numerical examples are explicitly tagged as authored fixtures. The inherited mathematical refinements are marked as master definitions rather than silently quoted as original source text.

Source integrity and release identity

SHA-256 identifies the exact available source bytes used by this edition: **A**

```
ba3dbe6f925c68ac0f237184b1a4f7770c4fe1d5c84b54bcb5a108dc37ec2ce3
```

F

```
1a91d9ce2af8c1ebdfc460913438cf127f1bde8be67c5a01c410976b5d7299ad
```

B

```
7c8e25b7269dc96b96e5e71afabb8d0019e905975f9959569ee4488b2c036ac8
```

O

```
90d8b3abadccce971917ea68e4f9be7b9d8d4a52ad7b700066649eb3ebc29732
```

D

```
05d20ca6b234950cf13abbc42257f60bebfa202e8e0c52c651acbdfebe240ec
```

aTOMos v3.6.1 — Total Integrated Engine

Foundational knowledge substrate S1 / Master M2

Tom Klootwijk NL200678942 10-07-1990

14 September 2026

The full core and the attached profile architecture are integrated in this volume. The ZIP contains the editable specification, declared foundation, complete local fixture, reference implementation, source map and reproducible blue-team evidence.